HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Managed a team of 5 engineers and 2 lab technicians, overseeing the planning, development, and delivery of camera tuning projects to meet automotive industry standards and regulatory requirements.
- Hosted weekly team reviews to align on project timing, assess maturity, and mitigate risks, ensuring timely achievement of development milestones for image quality KPIs.
- Translated OEM customer requirements for advanced driver-assistance systems into internal system and software specifications, actively coordinating with Systems Architects and Software Engineers.
- Drove continuous improvement in the camera tuning process by implementing automated testing and report generation for raw scene catalogues and HIL testing, optimizing image quality performance.
- Performed research into new vision technologies and approaches, focusing on protecting innovation and intellectual property through the patent process.
- Developed and executed training plans for team members, including 1:1 sessions and individual development plans, to ensure continued technical expertise in image processing and computer vision.
- Containerized embedded tuning software using Docker and deployed to an ARM-based embedded Linux platform, improving the consistency of testing across development and HIL environments.

SKILLS

Programming Languages – C++, Python, MATLAB, LabVIEW

Image Processing & Computer Vision – Image Signal Processing, Optics Fundamentals, Embedded Systems (ARM/DSP), PC-based Simulation Tools

Automotive Domain – Automotive Industry Experience, Camera Tuning, Image Quality KPIs, HIL Testing, NPI, Volume Production, Benchmarking

Development Process & Management – Agile Methodologies, Sprint Planning, Technical Leadership, Requirements Management, Risk Mitigation, Change Management, Team Development, IP Protection, Patent Process

Tools & Infrastructure – Jenkins, Git, Jira, Confluence, Docker, Linux, AWS (EC2/S3/Lambda/CloudWatch), PostgreSQL, MySQL, Oracle

PROJECTS

Embedded Camera Tuning Software Suite

- Developed a **Python**-based suite for camera tuning on embedded **ARM** platforms, integrating **MATLAB** scripts for image processing algorithms.
- Implemented automated testing workflows using **Jenkins** and **Docker** to validate image quality metrics against automotive regulations, reducing manual testing cycles by **40%**.
- Designed and maintained RESTful APIs using **FastAPI** for communication between the tuning application and embedded device firmware, ensuring robust data exchange for configuration parameters.

Image Quality KPI Automation Framework

- Engineered an automated framework in **C++** and **Python** to measure and report on key image quality KPIs, processing large datasets of raw scene images.
- Integrated with **AWS Lambda** and **S3** for scalable processing of image data and storage of generated reports, enabling real-time KPI tracking for development teams.
- Created interactive dashboards using **Power BI** to visualize tuning results and performance trends, facilitating data-driven decision-making for system architects and product managers.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)